CHAPTER SEVEN

CONCLUDING REMARKS

7.1 Thesis Summary

An ASR (automatic speech recognition) system was developed which incorporated a Fuzzy ART (Adaptive Resonance Theory) Map ANN (artificial neural network) as the component that classifies and recognizes speech samples presented to the system. The process of converting the spoken acoustic signal into a representation suitable for input to the recognizer component was described in terms of a four stage ASR process which included sound recording, spectral analysis, feature extraction and representation, and utterance classification.

Both microphone and telephone were used in the sound recording stage in order to measure and compare the system's performance on both transmission channels. The acoustic signals were digitized and converted into spectrogram form. Noise reduction, silence reduction and frequency amplitude normalization were carried out prior to tracking the first three speech formants. These formant were subsequently combined in a TFRV (tri formant ratio vector) representation that dramatically reduced the differences between the representation of the speech phrase due to speaker age and gender. The TFRV was then normalize to the range of fuzzy values (from zero to one) and LTW (linearly time warping) was performed to stretch or compress the TFRV to the size of the input layer of the Fuzzy ART Map ANN recognizer.

The combination of the TFRV speech signal representation and the Fuzzy ART Map ANN recognizer produced an ASR system capable of a speaker dependent recognition rate of 98.2% for microphone recorded speech and 91.5% for the same speech samples played over a noisy telephone line. The speaker independent rates were about 4% lower. Testing was

performed using the RMC Speech Database, which contained eight test phrases chosen from the literature for their diverse phonetic representation. The voices of the ten males and one female used in the testing, were recorded in a relatively noisy computer room over a period of four months in order to capture the variability of human speech.

A multi-layer feedforward ANN trained with the backpropagation algorithm (aided by a momentum term and an adaptive learning rate) was used as a benchmark in side by side comparison tests with the Fuzzy ART Map ANN. Its performace was approximately 5% lower than that of the Fuzzy ART Map ANN while the time required to train it was considerably longer.

7.2 Conclusion

The goal of this research was to investigate the applicability of the Fuzzy ART Map ANN to the task of Automatic Speech Recognition. **The development of the ASR system described in Chapter Five demonstrates that the Fuzzy ART Map ANN can be used to perform speech recognition, while the testing documented in Chapter Six establishes that the Fuzzy ART Map ANN can perform ASR more effectively than the ANN used in the benchmark tests** in the following ways:

- Its rapid learning capability **significantly reduced system training time** and produced **more efficient speech recognition**;

- Its localized (vector) data storage s**upports modular packaging and reconfiguration of active vocabulary**;

- Its use of a unit (ones) matrix to initialize its network memory **eliminates the uncertainty associated with random selection of starting weights** required by the benchmark ANN; and

- Unlike the benchmark ANN, whose fixed network size limited its storage capacity, the Fuzzy ART Map ANN **network expands in response to the need to learn "new" data** (i.e., significantly different from that which it already has stored in its memory).

7.3 Future Work

Improvements to the ASR system developed in this thesis stem from the error analysis in section 6.4.5. The development of an frequency amplitude normalization technique that adapts to the spectral attenuation of a given telephone connection would greatly improve the system's recognition performance in that area. Test scores may be improved by using on-line testing to take advantage of noise masks tailored to each speech sample. For telephone speech samples, temporal resolution could be improved to detect rapid changes in the speech signal through the use of a higher sampling rate, similar to that used for microphone recorded speech. Finally, the system could be made to execute faster if it was compiled as an executable program.

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